

Ankan Sarkar

B.Tech Student – Computer Science & Engineering (IoT) | Seeking AI/ML Engineering Intern | NLP

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EDUCATION

B.Tech – Computer Science & Engineering (IoT)

University of Engineering & Management, Kolkata

2024 – 2028

SGPA: 8.125 / 10

Higher Secondary (Class XII)

Durgapur Public School
CBSE Board

2024 · 68.4%

Secondary (Class X)

St. Michael's School
ICSE Board

2022 · 74%

TECHNICAL SKILLS

LANGUAGES

Python, C, Java

ML & DATA SCIENCE

Scikit-Learn, XGBoost,
LightGBM, Random Forest,
SVM, Logistic Regression,
Ensemble Learning, Pandas,
NumPy, SciPy, Matplotlib,
Seaborn, Feature
Engineering, Statistical
Analysis

DEEP LEARNING / NLP

PyTorch, TensorFlow, Keras,
DistilBERT, ResNet-18, DS-
CNN, ArcFace, TF-IDF, Text
Classification, Entropy-
Based Analysis

AUDIO AI

SpeechBrain, Librosa,
Torchaudio, Silero VAD, Mel
Spectrogram, Speaker
Verification, Cosine
Similarity

DEPLOYMENT & MLOPS

ONNX Runtime, Flask,
Gradio, INT8 Quantization,
REST APIs, Edge AI
Deployment

DATABASES & TOOLS

MongoDB, MySQL, Microsoft
Power BI, Google Sheets,
Microsoft Excel, Git/GitHub,
Jupyter, Google Colab, VS
Code, Conda, Overleaf

MODEL EVALUATION

Accuracy, Precision, Recall,
F1-Score, ROC-AUC,

PROFESSIONAL SUMMARY

Third-year B.Tech Computer Science (IoT) student at UEM Kolkata specializing in Artificial Intelligence, Machine Learning, and Data Science. Skilled in developing AI-powered cybersecurity and voice authentication systems using Python, Scikit-Learn, PyTorch, and NLP techniques. Passionate about building innovative AI solutions to address real-world challenges. .

EXPERIENCE

AI/ML Research Intern – Cybersecurity Division

May 2025 – Jul 2025

University of Engineering & Management (UEM), Kolkata

- Designed and deployed a **multi-class AI threat detection system** classifying spam, phishing, malware, botnet, DDoS, brute-force, and anomalous traffic with **98.23% overall accuracy**.
- Built end-to-end ML pipeline encompassing data preprocessing, feature engineering, feature selection, and hyperparameter optimization across **6+ threat categories**.
- Evaluated and benchmarked **five ML algorithms** — Random Forest, Extra Trees, LightGBM, Logistic Regression, and DistilBERT within a weighted voting ensemble framework.
- Integrated Explainable AI (XAI) to surface human-interpretable threat reasoning for security analysts and stakeholders.
- Authored IEEE-format research documentation, experimental analysis, and performance evaluation reports, contributing directly to a **peer-reviewed publication in 2026**.

PROJECTS

AI-Powered Cyber Threat Detection & Prevention System

May – Nov 2025

drive.google.com/...demo

Python · Scikit-Learn · PyTorch · DistilBERT · LightGBM · XGBoost · Extra Trees · TF-IDF · XAI · Pandas · NumPy · Matplotlib

- Built a multi-threat detection engine covering spam, phishing, malware, botnet, DDoS, and brute-force attacks — surpassing single-vector rule-based systems used in conventional security tooling.
- Designed an ensemble architecture fusing **transformer-based NLP (DistilBERT)** with gradient-boosted tree models via weighted voting, achieving **98.23% classification accuracy**.
- Implemented TF-IDF vectorization, entropy-based text analysis, and behavioral feature extraction to detect adversarial patterns invisible to standard classifiers.
- Validated model robustness via **Stratified K-Fold Cross Validation**; evaluated using ROC-AUC, precision-recall, and per-category confusion matrices.

OmniVoice – Edge-Based Voice Authentication System

May 2026 – Present

github.com/ASTER2005/Omni-Voice

Python · PyTorch · SpeechBrain · Silero VAD · DS-CNN · ResNet-18 · ArcFace · ONNX Runtime · INT8 Quantization · Librosa · Torchaudio

- Architected a fully offline, privacy-preserving voice authentication pipeline eliminating cloud dependency and the associated data-exposure risks of server-side voice processing.
- Achieved real-time Voice Activity Detection at **0.28 ms latency** and DS-CNN wake-word detection at **0.26 ms latency** using Silero VAD and custom feature extraction.
- Reduced speaker verification inference from **334 ms (PyTorch) to 1.00 ms (ONNX)** — a **99.7% latency reduction** via INT8 quantization and ONNX Runtime optimization.
- Implemented ResNet-18 + ArcFace with cosine similarity speaker matching; evaluated using FAR, FRR, EER, and min-DCF for deployment-grade biometric performance standards.

RESEARCH PUBLICATION

Real-Time Threat Intelligence and Automated Prevention Using Micro Intelligent System Technology and Explainable Tree-Based Models

Published · 2026 · AI-Driven Cybersecurity

ACHIEVEMENTS

- Finalist, Technologia Hackathon** – ranked among the top-performing teams nationally for developing an innovative technology solution under competitive time constraints.

Stratified K-Fold, FAR, FRR, EER, min-DCF

CYBERSECURITY AI

Threat Detection, Phishing, Malware, Botnet, DDoS, Brute-Force, Anomaly Detection, XAI

CERTIFICATIONS

Joy of Computing Using Python

NPTEL · IIT Madaras

INTERESTS

Generative AI, MLOps, Voice Biometrics, NLP, Predictive Analytics, Edge AI

- **Finalist, 4 Ideathons** – consistently advanced to final rounds across four ideation competitions, demonstrating cross-domain analytical thinking and solution design.
- **3rd Prize, Poster Presentation (×2)** – recognized twice for excellence in technical research visualization, communication clarity, and innovation in project demonstration.

LEADERSHIP & EXTRACURRICULAR ACTIVITIES

Logistics Lead – Awarded Best Working Member
Microsoft Student Society, UEM Kolkata

- Led logistics and execution for multiple student-led technical events and workshops; coordinated cross-functional teams and managed participant operations end-to-end.
- Awarded **Best Working Member** by the society for exceptional leadership, communication, and organizational contributions.

Treasurer & Sponsorship Volunteer

FusioniX'26 – 24-Hour National-Level Hackathon

- Managed full-cycle financial operations including budgeting, expense tracking, and financial planning for a large-scale national hackathon .
- Drove corporate outreach to secure sponsorships from technology companies and startups; authored professional sponsorship proposals and partnership agreements.